



Accuracy of Hanna marine nitrate HR checker in freshwater

A report prepared on behalf of Wye Salmon Association (WSA)

A quick “garage” evaluation of the Hanna marine nitrate checker for freshwater use, using a Hach standard solution.

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Introduction

Wye Alliance citizen scientists have historically used test strips for TON/nitrite monitoring, originally from Hach and more recently from WaterWorks. These are undoubtedly quick and simple to use; however, they do suffer from limited precision. A digital colorimeter would greatly improve precision, however the only available Hanna Checker falls within their “marine range” and has not previously been evaluated.

An obvious concern with an instrument with a “marine” designation is that calibration may be different when using it in freshwater. Hence, having obtained a Hanna marine nitrate HR checker for evaluation, the obvious first step was to test it using dilutions prepared from a Hach nitrate calibration solution.

The instrument

The instrument is very similar to the phosphate checker which we use, just a different colour. The operating procedure is also the same:

- Add sample to phial
- Zero the instrument
- Add reagent and shake for 2 minutes
- Return to instrument and start timer
- Obtain the reading

The principal difference is that the timer waits for 7 minutes rather than 2 minutes before taking the measurement.

As one might expect from Hanna, the unit of measure is unconventional. The reading is ppm (mg/l) as NO_3 , rather than the widely used mg/l as N. The conversion factor is 4.43.

Stated accuracy is $\pm 2\text{ppm} \pm 5\%$ of reading at 25°C (or $\pm 0.5\text{ppm} \pm 5\%$ of reading as N).

Range is 0 - 75 ppm NO_3 , which equates to 0 - 16.9 as mgN/l. This will cover most locations, most of the time but could over-range in point pollution situations.

Running costs are approximately £1 per measurement for reagent; the standard pack size is 25 sachets.

Method

As in the test strip trials, a standard solution of nitrate having a concentration of 100 mgN/l (Hach cat. 194749) was used to prepare 5 test solutions in separate beakers by stepwise dilution with DI water (1-5 $\mu\text{S}/\text{cm}$) using a measuring cylinder. Concentrations were chosen to cover the range of interest, as per table 1. Note that 20 mgN/l would over-range the meter, so is not used in the trial.

Concentration (mgN/l)	(As ppm NO_3)	Preparation
20		20% dilution of stock solution
10	44.3	50% dilution of 20 mgN/l
5	22.1	50% dilution of 10 mgN/l
2	8.9	40% dilution of 5 mgN/l
1	4.4	25% dilution of 2 mgN/l
0	0.0	100% DI water

Table 1: test solution preparation

At the time of testing, the solution temperatures were approximately 17°C .

Hanna readings were taken on each solution, with two repeats on 10mgN/l. Results are shown below.

Concentration (mgN/l)	(As ppm NO ₃)	Measured value (ppm NO ₃)
10	44.3	46.4
10	44.3	39.0
10	44.3	41.4
5	22.1	20.9
2	8.9	10.9
1	4.4	6.2
0	0.0	0.0

Table 2: Hanna checker readings on each solution

Figure 1 shows the plot, also including the error bands corresponding to the instrument specification.

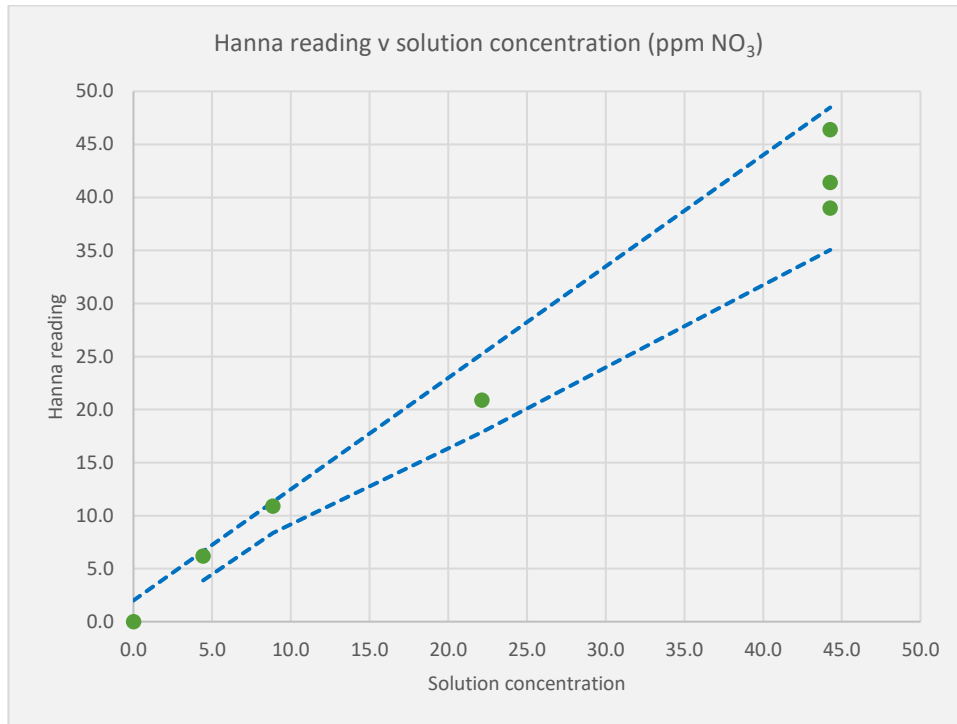


Figure 1; Hanna readings plotted against solution concentration (NB. stated as NO₃, not N!!). Blue lines indicate the range of the stated instrument accuracy (95%ile).

All readings fall within the range indicated by the stated accuracy of the instrument, without taking into consideration the additional uncertainties arising from the test solution and dilution process. The replicates on the 10 mgN/l solution suggest that most of the error, at least at the high end, is perhaps random rather than systematic.

Conclusion

- The Hanna marine nitrate HR checker performs to specification on non-saline solutions. There is no calibration change. It would therefore offer an additional option for monitoring nitrate in freshwater, where better precision is required.
- The range is suitable for most CS locations.
- The unit of measure is inconvenient and could cause confusion. If we pursued this more widely, a new EpiCollect field would be required.
- **Next steps** would be to put this unit out for field testing with one of our volunteers.

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